

We thank the editor and the reviewers for their comments.

Reviewer #1 (Formal Review for Authors (shown to authors)):

1. On lines 123-133 it states “While RCP8.5 is the most extreme emission scenario represented in the global climate models, it is a scenario becoming less plausible by 2100 and given recent shifts in global policies more moderate scenarios like RCP6 and RCP7 might be more plausible”

a. I recognize that you are using RCP 8.5 for your analysis so the statement above is contradictory to the rest of the paper. However, RCP 8.5 is the most likely scenario for temperature forcing. See [RCP8.5 tracks cumulative CO2 emissions](#), by [Christopher R. Schwalm et al.](#)

Thanks for pointing out this contradiction. We have modified the end of that paragraph to read:

“... The future expansion plans and loads used in this study are based on the rcp85hotter scenario in the TGW data (i.e., the hottest scenario). While RCP8.5 is the highest emission scenario represented in the global climate models, it is nonetheless a likely scenario over near- to midterm time horizons having good agreement with historical observations and future emissions under current policy (Schwalm et al., 2020).”

We would also like to point out that the methodology presented in the study does not strongly depend on which scenario may be more or less likely, the general conclusions are still the same in either scenario.

b. This is important for the conclusions in the study. If a zero-carbon electricity infrastructure is in place across the world by 2050, which the paper assumes, then the reduction in cooling from SO2 and other particulates will most likely increase the forcing to near the 8.5W/m2 level.

We have added the following sentence to capture this relevant observation:

“In addition, the reduction in the effects of cooling aerosols due to decarbonization may increase warming further (Dreyfus, 2022; Smil 2013) and positive feedback loops (Ripple, 2023; Moller, 2024). We note that the GODEEEP project and this study only assume decarbonization for the U.S. and not globally.”

c. However, it is more likely than not that significant emissions will still occur to 2050 as no government in the world is prepared to impose the scale of cutbacks that are

needed, so cumulative fossil fuel consumption to 2050 will continue to grow along the current trend and this will be exacerbated by increasing upwards pressure on demand.

This comment is addressed by the addition of the previous sentence regarding government policy.

d. Self-amplifying and interacting climate feedback mechanisms are now reducing the Earth ecosystem's abilities to remove CO2 from the atmosphere and in many cases are now adding CO2 such as through melting of the permafrost and heating of the surface waters of the oceans.

Thanks for bringing this up, we have added some reference regarding positive feedback looks.

“...and positive feedback loops (Ripple, 2023; Moller, 2024).”

e. So, it is important to defend your decision to use RCP8.5, otherwise your conclusions could be challenged, especially by those who are incentivized to diminish the problems of running an economy on renewables.

Thank you for the thoughtful series of comments. The additional text and references have addressed the likelihood of RCP8.5 as a climate scenario.

2. On lines 145-146 is states, “The GCAM-USA runs in this experiment used the Shared Socioeconomic Pathways (SSP) 2 scenario for socioeconomic and population forcing and the rcp85hotter TGW climate scenario.”

a. While SSP 2 is the steady state, observational evidence from across the world shows Regional Rivalry is now very much the dominant situation. Notably, since the SSPs were defined, we have the Ukraine War, the Middle Eastern Wars, and large-scale illegal immigrant border invasions in the USA and EU, prospective cold war confrontation with China, and conflicts in Africa.

We agree that SSP3 would have been another reasonable choice. The choice was made early in the project to use SSP2 as opposed to SSP5 due to the industry focus where SSP5 was looked on with skepticism.

If we had gone with SSP3, we expect the larger population/economic growth would impact the amount of capacity coming out of GCAM and potentially the resource mix, which in

turn would impact siting decisions. More directly, areas that see large population growth would have more excluded area for siting.

It is difficult to quantify the exact effect without running the entire model chain for the capacity expansion, which was out of the scope of this study.

b. This is significant as energy droughts will be more impactful in an SSP3 world than a SSP2 world.

While we would expect the larger population to require increased wind and solar buildout, as shown in the manuscript, this can have a paradoxical effect of lowering the changes for energy droughts at the BA level. This is because with a larger number of wind and solar plants, the conditions required for an entire BA region to be in drought are stricter. Although we do recognize that the absolute magnitude of drought (MWh deficit below threshold) would be greater with more plants.

3. On lines 279 the definition is given, “Our definition of compound VRE drought severity is the total energy deficit below the 10th percentile drought threshold”

a. I don’t understand what this is telling me. I am assuming that the energy deficit is the difference between demand and supply, perhaps given by the formula that I infer from line 263:

$$\text{energy deficit} = \frac{\text{energy supply} - \text{energy demand}}{\text{total energy supply}}$$

We do not have reliable data regarding the full energy mix of each BA, so this definition of severity is a proxy for the full definition which would be similar to what you provided above. That said, the cumulative deficit below a particular drought threshold is a commonly used measurement for severity used in the hydrologic literature.

To improve the clarity of the definition of compound drought severity we have added the following sentence to the manuscript:

“While this definition is a proxy for the true energy deficit which requires data on the full energy mix at every BA, it is a commonly used metric (Allen 2023).”

b. The formula for energy deficit needs clarification. I have assumed total energy supply in the denominator. However, this is a function of the generating conditions, and it is not clear if a nominal value is used in the calculations or if the value is calculated

dynamically based on the climate simulations. For example, in a sunny and windy day the energy supply would increase.

See the previous comment.

c. Alternatively, the denominator might be total energy demand, which will also be a function of generating conditions, for example, air conditioning demand would increase in very hot weather conditions.

The denominator as we used it in the paper is simply a normalizing factor so that the severity calculations can be compared across different BAs. We used the maximum combined generation from wind and solar. Total energy demand would have also worked but we do not have reliable data for every BA.

d. On this basis, what is the data in Figure 3 that is used in the box plots? Is it the number of days per simulation that an energy drought occurs, or the number of days in a single simulation above the 10% threshold, or the total number of energy deficit events over a single simulation? A better description of this is needed and better metrics.

Figure 3 is the calculated severity for each BA in each infrastructure year. The boxplots represent the range of weather years. Severity is calculated by calculating the cumulative energy deficit for both wind and solar during a compound drought, and then dividing that value by the maximum combined wind and solar value for the entire period.

The following sentence in the manuscript has been improved for clarity:

“Drought severity is measured as the cumulative energy deficit for both wind and solar below the drought threshold during a drought event, which can be expressed in megawatt-hours (MWh).”

e. There is nothing to justify the 10% cut off level for an energy drought, and it looks rather arbitrary. Is this selected because a 10% shortfall in energy supply is considered manageable through initiatives to manage demand or is it considered that 10% is the accuracy of the simulation? This is where the SSP2 v SSP3 issues becomes of more importance. In a scenario of Region Rivalry, it is more likely that a loss of energy supply would have a more significant impact, for example, a nation would not wish to lose any power supply necessary for the operation of its military industrial complex or its emergency facilities at times of crisis. In this case 10% may be far too high.

You are correct, the threshold is arbitrary and in practice should be set for each BA and generating resource individually. While the threshold may capture some less extreme compound droughts, it is sufficient to represent a range of conditions for this study. We have added the following sentence to the manuscript to address this concern.

Note that this threshold is arbitrary and in practice should be defined for each BA and generating resource based on regionally specific impacts.

f. It would be more useful to have the raw data displayed with confidence intervals. So perhaps show graphically the trend from now to 2050 of the total cumulated energy deficits per season and the number of droughts and the size of the droughts. I would image this data is available from the simulations that have been done.

We do have the raw data from the generation simulations and the unnormalized energy drought values. These data are available on Zenodo (links in manuscript). It is challenging to represent all the data in a concise way due the very different scales involved from different BAs which is why we opted for a normalized comparison. A full uncertainty quantification is left for future work.

Reviewer #2 (Formal Review for Authors (shown to authors)):

This paper evaluates the compound wind and solar energy droughts for their potential impact on grid reliability in a future decarbonized or low-carbon grid. It considers the future climate scenarios, more importantly, and infrastructure scenarios, which makes it a good application from climate science to real policy. This paper is clear in significance, methods, discussion, and writing to make the reading process enjoyable. I think this manuscript is almost ready to be published.

Minor feedback is to further clarify the definition: do variable renewable energy (VRE) droughts mean prolonged periods of low wind and solar generation? If yes, then what is the definition of compound VRE droughts? Or is the definition of VRE droughts "prolonged periods of low wind or solar generation"? Please clarify these definitions in the paper.

Thank you for the comment.

Yes, VRE droughts mean periods of low wind, solar, or hydro generation, but they may last any amount of time from hours to months depending on the data used to define the droughts. The drought actually occurs when the generation falls below some predefined threshold, we used 10th percentile in this study. A compound drought occurs when 2 or more renewable generating sources are in drought conditions simultaneously. This has been clarified in the paper by adding this sentence to the introduction:

“VRE droughts may last for hours to months depending on the data used to define the droughts. The drought is said to occur when the generation falls below some predefined

threshold. A compound drought occurs when 2 or more renewable generating sources are in drought conditions simultaneously.”